A ReasonScore Details

A.1 Reasoning Vocabulary

In Fig. 5, we show the complete list of words from the *reasoning* vocabulary $\mathcal{R}$ that we obtain in Sec. 3.1. For clarity, we list only the lowercase forms without spaces (e.g. "alternatively"). However, in our implementation, we track multiple forms of each word, including capitalized ("Alternatively") and space-prefixed variants (" alternatively", " Alternatively"), as the tokenizer can assign different tokens for each of the forms.

Our vocabulary includes several words that require additional clarification. Concretely, we retain "but" despite exceeding the frequency threshold, as a strong reasoning indicator, as shown in prior research. Empirical analysis further confirmed its association with reasoning transitions. We include "let me" as a phrase rather than "let" alone, since model consistently pair these tokens as a single reasoning unit. For "but" and "another", we use only capitalized forms, as lowercase variants appear predominantly in non-reasoning contexts while capitalized forms mark reasoning transitions.

```
["alternatively", "hmm", "maybe", "wait",
"perhaps", "let me", "therefore", "however",
"but", "another"]
```

Figure 5: The complete list of words from the *reasoning* vocabulary $\mathcal{R}$ in the lowercase and without spaces form.

A.2 Importance of Reasoning Words

To address whether our selected reasoning words truly reflect model's reasoning capabilities or merely act as superficial linguistic markers, we conduct an ablation study that measures the importance of the reasoning vocabulary on model performance. Specifically, during the model generation, we **ban** words from our reasoning vocabulary so the model can't output them during its thinking process. We then measure the performance of this setup using pass@1 metric on AIME-2024, MATH-500, and GPQA Diamond. To isolate the impact of reasoning words from the general effect of banning any words, we conducted an additional experiment where we banned an equal number of words, with each word *randomly* sampled from the p_{think} distribution (Sec. 3.1). For the *random words* experiment, we repeat it 10 times with different sets of words and report average pass@1.

Word-ban	AIME 2024		MATH-500		GPQA Diamond	
	pass@1	tokens (K)	pass@1	tokens (K)	pass@1	tokens (K)
Default	53.3	12.4	93.2	3.9	50.0	7.9
Reason	**20.0**	**9.1**	**86.6**	**2.5**	**41.9**	**5.7**
Random	48.6	13.3	89.4	4.2	48.5	8.9

Table 2: Performance and average number of output tokens (K) under word-ban experiments. **Default**: no banned words; **Reason**: ban the reasoning vocabulary; **Random**: ban random words.

The results, presented in Tab. 2, demonstrate that banning *reasoning* words significantly decreases both the performance and the reasoning depth (average number of output tokens). In contrast, banning random words produces no statistically significant differences from the baseline. This provides empirical evidence that these reasoning words play a functional role in the model's reasoning process.

A.3 ReasonScore Distribution

Fig. 6 shows the ReasonScore values sorted in decreasing order of all SAE features for DEEPSEEK-R1-LLAMA-8B. We select the 0.997-th quantile as a cutoff, yielding approximately 200 features. While the plot shows the ReasonScore continues to decay below this threshold rather than reaching a plateau, this amount is feasible to analyze manually and contains all the most important features as judged by our metric.

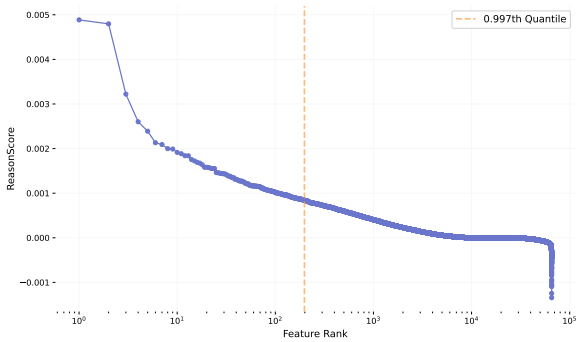

Figure 6: Distribution of ReasonScore values across all SAE features for DEEPSEEK-R1-LLAMA-8B, sorted in decreasing order.

B Interpretability Details

B.1 Feature Interfaces

Figs. 8,9 display additional activation patterns for features we found during manual interpretation (Sec. 4.2), highlighting tokens where each feature most strongly activates.

B.2 Automatic Interpretation Details

To interpret each feature, we randomly selected 30 examples from the OPENTHOUGHTS-114K dataset. The selection was stratified across different domains to ensure a balanced representation of various reasoning types and application contexts. The automatic feature interpretation followed a two-step pipeline. First, we used GPT-4O to generate free-form descriptions of the inferred function of each feature. This step involved detailed prompting, requesting the model to explain what the feature likely encodes and provide supporting examples.

After generating individual descriptions, we cluster reasoning-related features based on their possible functions and behavioral patterns. To support this process, we provided GPT-4O with a list of existing features, accompanied by a description of its possible function and observed steering behavior. The model was asked to identify recurring patterns and group similar features accordingly. To ensure accuracy, the results were then manually reviewed and validated. Table 4 presents the resulting feature groups, including categories (reasoning depth and thoroughness, self-correction and backtracking, and others), along with descriptions of their roles and effects. In some cases, even features grouped together based on shared function exhibited subtle differences in how they influenced responses; for instance, among features encouraging structural organization, one may focus on logical flow and paragraphing, while another influences transitions between argument steps. Additionally, features often demonstrated overlapping effects with other groups or influenced aspects beyond reasoning alone. For example, affecting the stylistic tone or structure of the output. This suggests that certain features may play a broader role across different types of reasoning and expression, rather than being confined to a single function.

C Examples of Feature Steering

Tab. 5 show the example of model's thinking process on a "how many r's in the word strawberry" question with and without steering. Tabs. 6,7,8 show the examples of model's thinking processes on reasoning-related benchmarks with and without steering.

Layer	Reasoning Features			Random Features		
	diff	maybe	same	diff	maybe	same
13	52.5	14.0	**33.5**	76.0	9.0	15.0
15	43.5	17.0	**39.5**	63.0	13.0	24.0
17	27.0	17.0	**56.0**	49.0	13.0	38.0
21	17.5	23.5	**59.0**	50.0	9.0	41.0
23	17.5	22.0	**60.5**	54.0	15.0	31.0

Table 3: Cross-layer matching results for reasoning features and 100 random features from layer 19-th SAE. All values are in percentages (%).

D Generalization Experiments

D.1 Reasoning Capabilities Across Layers

We perform additional experiments by analyzing the existence of similar *reasoning* features across layers. We train multiple SAEs on activations from layers $\{13, 15, 17, 21, 23\}$ of the DEEPSEEK-R1-LLAMA-8B model. To match the features between different SAEs, we use the approach proposed in (Balagansky et al., 2024). For each layer, we match all features from our original 19-th layer SAE with all features in the considered layer. Formally, we aim to find a permutation matrix that aligns the features of layer A with those of layer B. This is done by minimizing the MSE between the decoder weights:

$$\begin{aligned} P^{(A\to B)} &= \arg\min_P \sum_{i=1}^{n} ||W^{(A)}_{\text{dec},i} - PW^{(B)}_{\text{dec},i}||^2 \\ &= \arg\max_P \langle P, (W^{(A)}_{\text{dec}})^T W^{(B)}_{\text{dec}} \rangle_F. \end{aligned} \tag{8}$$

After matching each i-th feature from layer 19 with some other j-th SAE feature from a different layer, we take the subset corresponding to *reasoning* features ($\mathcal{F}_{\mathcal{R}}$) and their corresponding matched subset. We evaluate the similarity of a pair of matched features (i, j) using a *Matching Score*, which is defined as the cosine similarity between feature activations across a batch of samples from OPENTHOUGHTS-114K:

$$\text{MS}(i,j) = \frac{\sum_{x\in D} f_i(x) \cdot f_j(x)}{\sqrt{\sum_{x\in D} f_i^2(x)} \cdot \sqrt{\sum_{x\in D} f_j^2(x)}}. \tag{9}$$

Here, D is a subset of token activations from OPENTHOUGHTS-114K, $|D| = 10$M tokens.

We consider i-th reasoning feature to be presented in another layer as j-th feature if $\text{MS}(i,j) > 0.7$ (**same**), undefined if $0.5 < \text{MS}(i,j) \leq 0.7$ (**maybe**), and absent if $\text{MS}(i,j) \leq 0.5$ (**diff**).

```
Feature #19371
, output NO.\n\nBut how to efficiently implement this.\n\n
 Return moving_avg\n\nBut what if the CSV has different
 to the count.\n\nBut how to compute the intersection.\n\n
 get the count.\n\nBut how to build these hash maps
Feature #4646
 more.\n\nAlternatively, maybe there's a non-symmetric
 insight.\n\nWait, maybe I need to assume that the
 n.\n\nWait, maybe there's a different way to
als are perpendicular, maybe there's a simpler formula.
Feature #13448
 is considered known. But maybe in the context of the
 format, it's possible that the expected answer is \
No}.\n\nAlternatively, maybe the problem expects the answer
 problem seems more elementary, maybe expecting Riemann.\n\n
Feature #46772
value.\n\nWait, perhaps we can think of the total
structure.\n\nWait, perhaps we can find that the updates
problem.\n\nWait, perhaps the minimal sum is not achieved
enough.\n\nAlternatively, perhaps we can use a heap,
```

Figure 7: Top-activating examples from the manually verified set of Nemotron features.

We present the results in Tab. 3, along with the matching scores for the 100 randomly sampled features from 19-th layer SAE to establish a baseline for comparison. Our analysis reveals that similar reasoning features exist across model layers (as judged by the *Matching Score*). For example, 39.5% of the features from layer 19 have close matches (MS > 0.7) in layer 15, significantly higher than the 24.0% matching rate for randomly chosen features.

D.2 Reasoning Capabilities Across Architectures

We train the SAE on the 19-th layer of a widely adopted open-source family of reasoning models, Nemotron (Bercovich et al., 2025). Specifically, we use the LLAMA-3.1-NEMOTRON-NANO-8B-V1 model and utilize its original training dataset to train our SAE. Otherwise, we follow the same experimental setup as in Sec. 4.1 of the main paper. We compute the `ReasonScore` for all features and select the top-200 as judged by the `ReasonScore`.

Following Sec. 4.2 of the main paper, we perform a manual interpretation of these features. We find the Nemotron features that have activation patterns similar to the ones in the DEEPSEEK-R1-LLAMA-8B model. In Fig. 7, we provide interfaces of features representing model's uncertainty (#46772), exploration (#13448), and reflection (#4646, #19371). These results indicate that reasoning features emerge across distinct families of reasoning LLMs.